



Human health risk assessment (HHRA) of fluoride and nitrate using pollution index of groundwater (PIG) in and around hard rock terrain of Cuddapah, A.P. South India

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ABSTRACT

Groundwater is the most common source of drinking and irrigation in the world. As a result of human activities and natural processes, nitrate levels in most groundwater supplies in arid and semi-arid areas have risen in recent decades. As a result of human activities and natural processes, the levels of fluoride and nitrate in most groundwater supplies in arid and semi-arid areas have increased in recent decades. Thirty groundwater samples were taken from bore wells to learn about the contaminated situations of groundwater fluoride and nitrate in sections of Cuddapah, South India, and to determine the likely harm to the health of adults, children, and infants. In the study villages' wells, fluoride concentrations range from 0.1–3.2 mg/L, with 40% of groundwater exceeding fluoride guidelines (1.5 mg/L) and Nitrate concentrations range from 23.2–110.8 mg/L, with 86.6% of groundwater exceeding nitrate guidelines (45 mg/L). 96% of groundwater exceeds the allowable THI values (>1) for adults (0.74–4.02), children (0.75–4.08), and infants (0.57–3.10). According to the PIG categorization, 23% of total groundwater samples fall into the low pollution zone (PIG: 1.38) whereas 47% of samples fall into the moderate pollution zone (1.83) and 30% of groundwater samples fall into the high pollution zone (2.20). Hence, appropriate preventative steps must be adopted to reduce the health risk in this area.

1. Introduction

Groundwater constitutes an important onset of fresh water for drinking and agriculture in the world's arid and semi-arid regions. In the world, more than 1.5 billion people entrust on groundwater for their drinking water. Groundwater is the primary source for meeting the needs of rural residents for daily activities, particularly in a developing country like India [1,2]. In dry and semi-dry parts of various countries, groundwater is the primary replenishable supply of fresh water for residential, agriculture, and industrial needs. Drinking water accounts for 65% of groundwater use globally, farming and animal feeding accounts for 20%, while industry and mining accounts for roughly 15% of groundwater use [3]. Quality of groundwater in the rain shadow zone of semi-arid regions is characteristically a function of regional geo-hydrology, lithology, evapotranspiration, rainfall quality, and quantity, as well as host rock reactions with precipitation, inorganic compound solubilities, aquifer mineralogy, and weathering characteristics.

Anthropogenic factors such as industrial processes and wastes, the release of fertilizers, herbicides, and animal manure lower beneficial uses of groundwater resources [4–6]. Human health and water quality are inextricably linked. Water pollution by fluoride and nitrate is a serious public health concern around the world for the past two decades. These research works were carried out in arid and semi-arid locations everywhere in the globe due to their importance [7–13].

Fluoride levels are higher in greater than 23 nations globally, especially in India, China, Central Africa, and South America. Several groundwater quality studies with fluoride were conducted in India's arid and semi-arid regions. Even though fluoride is necessary for the natural mineralization of bones and the creation of dental enamel in modest amounts, higher concentration levels (>1.5 mg/L) produce dental fluorosis or mottled enamel, and elevated fluoride concentrations (>3.0 mg/L) can cause skeletal fluorosis [14–21]. Fluoride concentrations in volcanic rocks (viz., granites, gneisses, and basalts) range from 300 to

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